Instructions to the candidates:

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Figures to the right side indicate full marks.
3. Assume suitable data, if necessary.
4. Neat diagrams must be drawn wherever necessary.

Unit III --- Network Layer

Q1) a) Differentiate between Circuit Switching and Packet Switching. [6]

b) Write a short note on RIP (Routing Information Protocol). [6]

c) Given the IP address 192.168.5.71 / 26, find: i) Subnet mask ii) First usable IP address in the subnet iii) Last usable IP address in the subnet iv) Broadcast address [6]

OR

Q2) a) Draw and explain the IPv6 header format. Discuss the improvements of IPv6 over IPv4.

b) Write a short note on BGP (Border Gateway Protocol). [6][6]

c) List and explain the functions of the Network Layer. [6]

Unit IV --- Transport Layer

Q3) a) Draw and explain the TCP header format. Explain the significance of each field. [6]

b) List and explain the transport layer services. [6]

c) Given the UDP hexadecimal dump: `e2 a7 00 0d 00 20 74 9e 0e ff 00 00 00 01 00 00 00`, extract
in decimal: i) Source port number ii) Destination port number iii) Total length of user datagram [6]

OR

Q4) a) Draw and explain the UDP header format. [6]

b) What is a socket? Explain different types of sockets. Describe the socket functions used in connection-oriented services with a diagram. [6]

c) Explain the SCTP protocol in detail, including its features and multihoming concept. [6]

Unit V --- Application Layer

Q5) a) What is HTTP? Explain HTTP request and response messages with formats. Differentiate between HTTP/1.1 and HTTP/2. [9]

b) Write short notes on SMTP and MIME. Explain how email delivery works using SMTP, POP3, and
webmail. [9]

OR

Q6) a) What is DHCP? Explain the DHCP working with a client state diagram. [9]

b) Write short notes on: i) DNS (Domain Name System) --- explain recursive and iterative query resolution ii) FTP (File Transfer Protocol) --- control connection vs data connection [9]

Unit VI --- Security

1

Q7) a) Draw and explain the ITU-T X.800 Security Architecture for OSI. [6]

b) Write a short note on HTTPS and its role in secure web communication. [6]

c) Write a short note on Intrusion Detection Systems (IDS) --- types and operational mechanisms. [5]

OR

Q8) a) Differentiate between Symmetric and Asymmetric Key Cryptography with examples. [6]

b) Explain SSL (Secure Socket Layer) in detail --- protocol stack, handshake, and record protocol. [6]

c) Write a short note on Firewalls: types, architecture, and advantages. [5]

Examiner Commentary

This paper follows the standard SPPU CN pattern --- short note-heavy with numerical subnetting and UDP header parsing. Q1(c) tests practical IPv4 subnetting. Q3(c) requires parsing a real UDP dump. Each OR pair covers one unit with balanced theory and numerical components. Security questions (Q7/Q8) include both architectural (X.800) and applied (SSL, Firewalls) knowledge.

2